

Title :- Introduction to Mlops, Comparison of Software Development, Devops & Mlops, Challenges in production ML system, Survey of Mlops Tools & Technologies, Preparation of Mlops Workflow Diagram & Report.

Aim:- To study the fundamentals of Mlops, compare software development, Devops, Mlops, understand the Mlops lifecycle & its key compo., identify challenges in production ML system surveys Mlops tools & technologies & prepare on Mlops workflow & report.

Objectives-

1. Understand concepts & importance of Mlops
2. Compare software development, Devops & Mlops

Introduction-

Mlops stands for Machine learning operations it is combination of ML & Devops Practices used to develop, deploy monitor & maintain ML models in production.

It provides a systematic approach to Automate & manages Activities.

Explanation-

- Software development mainly focuses on creating software applⁿ.
- Devops combines development & operations to automate software development, testing

deployment & Monitoring.

- Mlops applies Devops principles to ML system while additionally managing data models experiment & retraining.

Mlops Lifecycle :-

Stage 1 - Data collection -

Data is collected from different source such as databases, API's, files, sensors

Stage 2 - Data preparation -

collected data is cleaned & transformed.

Stage 3 - Model development -

A ML algo. is selected & trained using prepared data.

Stage 4 - Model Evaluation -

The model is evaluated using suitable performance metrics. Accuracy, Recall.

Stage 5 - Model Deployment -

The trained model is deployed into a production environment so that users or app'n can access it.

Stage 6 - Model monitoring -

model continuously monitored For: Accuracy

Stage 7 - Model Retraining -

model performance decreases new data.

Key Components -

- 1) Data management - Managing, storing
- 2) Model development - Develop & train
- 3) Experiment Tracking - Recording model parameter
- 4) Version control - Maintains diff. version
- 5) Model Registry - Managing trained models
- 6) CI/CD - testing, building & process
- 7) Model Monitoring - performance & health

Conclusion -

Mlops provides a structured & auto approach for managing the complete lifecycle of ML deployment models, It combines ML, Software development, Devops, Automation deployment & monitoring to make ML system reliable, scalable & maintainable in production.